

From Neural Networks to Agents

Zhiwei Li · AI Engineer

lzwjava@gmail.com · github.com/lzwjava · lzwjava.github.io

Ability Check and Factual Check

Before we start, a factual/trust check on this deck:

- **Made with Claude Code & Marp** — slides are AI-assisted, not hand-written
- Built from my public AI response notes — lzwjava.github.io/notes-en.html
- **Prompted by me, verified largely by me** — but I am the bottleneck, not an infallible reviewer
- I honestly understand **~50%** of the content in this deck well; the rest I can explain but not derive cold
- Compared to **Andrej Karpathy**, my grasp of these topics is maybe **~30%** — he can build GPT from scratch on a whiteboard; I still reach for the code

Treat this as a **learner's map**, not an authority. Verify important things before you depend on it.

About Me

- **11 years** in software engineering — mobile, backend, full-stack, AI
- Currently at **a global IT services company** — contractor at **a global bank** — backend + AI-assisted development
- ~**1 billion tokens** consumed by OpenRouter & other providers in past year
- Training GPT-2 with **nanoGPT** on **H200 GPU**
- Previously: contractor at DBS Bank & HSBC PayMe, startup founder (30k users)
- **AI/LLM:** PyTorch, nanoGPT, Claude Code, OpenRouter, Copilot, llama.cpp
- **Backend:** Java, Spring Boot, Python | **Cloud:** AWS, Azure | **Mobile:** iOS, Android
- **2011 NOIP Guangdong** — 1st prize (Guangzhou, round 1), advanced to round 2, ~top 300
- **Born 1995** | **IELTS 6.5** (Reading 8.5)

AI Journey

- Completed **ML Specialization** & **Deep Learning Specialization** (Andrew Ng)
- Training GPT-2 124M with **nanoGPT** on H200 GPU
- ~**1 billion tokens** consumed via OpenRouter & other providers
- Built custom agents, prompt pipelines, automation tools
- Experimented with llama.cpp, embeddings, MMLU benchmark
- Developing autonomous workflows with **Claude Code** & **OpenClaw**

My Path to Understanding Transformers

- First read about **K, Q, V** mechanism around end of **2023** — didn't understand much
- Read *The Illustrated Transformer*, watched Karpathy, Umar Jamil, StatQuest
- Replicated neural network for handwritten digits **from scratch** — that's where real understanding began
- By mid-**2025**, transformers finally clicked — after **~2 years** of mulling
- Key insight: you don't get it the first time — the brain does its work over time

"If one can write it from scratch without copying any code, one understands very well."

Blog post: [Neural Network, Transformer and GPT](#)

GTY — Great Teacher Yin

Yin Wang (yinwang.org) — 140k followers on Weibo, independent thinker, CS researcher.

- Followed him since **2013** — he taught me how to think from first principles
- His professor: **Daniel P. Friedman** (Indiana University, born 1944)
- Yin Wang calls ML "**differentiable computing**" — it's just calculus
- His vision restoration method inspired my **3-year myopia experiment**
- Subscribed to his Substack since 2022

"He not only taught me knowledge, but truly showed me how to think. He passed on the technique of catching fish from Daniel P. Friedman to me."

Blog post: [GTY - Great Teacher Yin](#)

"Differentiable Computing"

"Machine learning is really useful, one might even say beautiful theory, because it is simply **calculus after a makeover**! It is the old and great theory of Newton, Leibniz, in a simpler, elegant and powerful form."

"There is no 'intelligence' in artificial intelligence, no 'neural' in neural network, no 'learning' in machine learning. What really works in this field is called '**calculus**'."

— **Yin Wang**

See through the buzzwords to the math underneath.

Learning Philosophy

Inspired by **Yin Wang** and **Andrej Karpathy**

- **Print variables** to understand — not just read about it
- **Read real code** (nanoGPT, not just textbooks)
- **Train small models** yourself
- **Build systems**, not just theory
- Iterate like training an LLM — you don't get it right the first time

Learning Path Overview

Foundations (1–4): Neural net from scratch → PyTorch → language model

GPT (5–7): Transformers → train nanoGPT → scale up training

Agents (8–10): Chat model → coding agent → personal AI system

From math to GPT to AI system. That's the path.

Neural Networks from First Principles

Reference: Michael Nielsen — [Neural Networks and Deep Learning](#)

Understand what a neural network really computes.

- Input: 784 pixels (28×28 handwritten digit) → Output: 10 neurons (digits 0–9)

- Each neuron: **weighted sum + bias + activation function**

- $$a^l = \sigma(w^l a^{l-1} + b^l)$$

- Sigmoid maps any value to 0–1: $\sigma(z) = \frac{1}{1+e^{-z}}$

For a 2-layer network: **784 × 10 weights + 10 biases** to learn.

Read Alongside — neuralnetworksanddeeplearning.com

Open the site in your browser and read **chapter by chapter**. This talk walks alongside it.

- **Ch 1:** Using neural nets to recognize handwritten digits
- **Ch 2:** How the backpropagation algorithm works
- **Ch 3:** Improving the way neural networks learn
- **Ch 4:** A visual proof that neural nets can compute any function
- **Ch 5:** Why are deep neural networks hard to train?
- **Ch 6:** Deep learning

Free online. Interactive diagrams. Python code on GitHub: [mnielsen/neural-networks-and-deep-learning](https://github.com/mnielsen/neural-networks-and-deep-learning)

Backpropagation — How Networks Learn

1. **Input:** set activation a^1 for input layer
2. **Feedforward:** compute $z^l = w^l a^{l-1} + b^l$, then $a^l = \sigma(z^l)$
3. **Output error:** $\delta^L = \nabla_a C \odot \sigma'(z^L)$
4. **Backpropagate:** $\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$
5. **Update:** $w \rightarrow w - \eta \frac{\partial C}{\partial w}$, $b \rightarrow b - \eta \frac{\partial C}{\partial b}$

Practice — Build a Neural Network from Scratch

Replicate the MNIST digit recognizer in pure Python + NumPy:

```
def __init__(self, sizes):          # e.g. [784, 30, 10]
    self.weights = [np.random.randn(y, x)
                     for x, y in zip(sizes[:-1], sizes[1:])]
    self.biases = [np.random.randn(y, 1) for y in sizes[1:]]
```

- Print every variable shape: `weights[0].shape = (30, 784)`
- Watch accuracy climb: 10% → 50% → 95% over epochs
- **If you can write it from scratch, you understand it**

After this, you understand every number inside a neural network.

Dimensions — The Key to Understanding

Know the **shape** of every variable in a neural network:

Variable	Shape	Meaning
Input <code>x</code>	<code>(784, 1)</code>	28×28 pixels flattened
<code>weights[0]</code>	<code>(30, 784)</code>	Layer 1 → Layer 2 connections
<code>biases[0]</code>	<code>(30, 1)</code>	One bias per hidden neuron
<code>z = w · a + b</code>	<code>(30, 1)</code>	Weighted sum before activation
<code>a = σ(z)</code>	<code>(30, 1)</code>	Activated hidden layer
<code>weights[1]</code>	<code>(10, 30)</code>	Layer 2 → Output connections
Output	<code>(10, 1)</code>	Confidence for digits 0–9

Matrix multiply: `(10, 30) × (30, 1) = (10, 1)` — dimensions must align.

"It just does the differentiable computation. Know the dimensions of every

From Neural Networks to Deep Learning

How deep learning actually trains.

- Gradient descent, learning rate, convergence
- Overfitting vs generalization
- Regularization, dropout, batch/mini-batch/SGD

Practice: Train a 3-layer classifier. Visualize the loss curve. Implement dropout manually.

Language Modeling Fundamentals

Understand what GPT predicts.

- Tokenization (BPE), n-gram models
- RNN/LSTM intuition
- Next token prediction, cross-entropy for language

Practice: Build a character-level language model. Train on tiny Shakespeare. Generate text.

PyTorch Minimal Framework

Read any PyTorch model with confidence.

- Tensor fundamentals, autograd
- `nn.Module` design, optimizer mechanics
- Dataset and DataLoader

Practice: Rebuild MLP in PyTorch. Train a CIFAR classifier. Inspect gradients.

GPT Architecture — The Big Picture

Reference: Andrej Karpathy — [Let's build GPT: from scratch, in code](#)

Input Text

- Tokenizer (text → token IDs)
- Token Embedding + Positional Encoding
- $N \times$ Transformer Blocks
 - (Self-Attention → Feed-Forward → LayerNorm)
- Linear → Softmax
- Next Token Prediction

The entire model learns to predict: **given these tokens, what comes next?**

Tokenizer — Text to Numbers

Before the model sees anything, text must become numbers.

- **Character-level:** each character is a token — simple but slow
- **BPE (Byte Pair Encoding):** merge frequent character pairs iteratively
 - "learning" → ["learn", "ing"] → [4821, 278]
- GPT-2 uses ~**50,257** tokens; GPT-4 uses ~**100k**
- Tokenizer is trained **separately** before the model

```
"The pizza came out" → [464, 13293, 1625, 503]
```

Different tokenizers produce different token IDs — changing the tokenizer changes everything.

Word Embedding — Tokens to Vectors

Each token ID maps to a **learned vector** (e.g., 768 dimensions in GPT-2).

- token 464 ("The") → [0.12, -0.34, 0.56, ..., 0.08]
- These vectors are **not hand-crafted** — they're learned during training
- Similar words end up with similar vectors

Positional encoding is added so the model knows word order:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d})$$

Without position info, "dog bites man" = "man bites dog" to the model.

Self-Attention — How Words Relate

| "The pizza came out of the oven and **it** tasted good."

How does the model know "it" refers to "pizza" and not "oven"?

Self-attention computes a similarity score between every pair of tokens:

1. Each token produces a **Query**, **Key**, and **Value** vector
2. Dot-product $\text{Query} \times \text{Key} = \text{attention score (how relevant?)}$
3. Softmax normalizes scores to weights (0 to 1)
4. Weighted sum of Values = output for that token

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Q, K, V — Intuition

Think of it like a **search engine** inside the model:

Component	Role	Analogy
Q (Query)	What am I looking for?	Your search query
K (Key)	What do I contain?	Page titles / descriptions
V (Value)	What's my actual content?	Page content returned

For the word "it":

- **Q** asks: "what noun am I referring to?"
- **K** of "pizza" answers: "I'm a noun, a food, the subject"
- **K** of "oven" answers: "I'm a noun, an appliance, inside a prepositional phrase"
- Attention score for "pizza" > "oven" → **V** of "pizza" contributes more

Q, K, V — Dimensions in Attention

For a sequence of **T** tokens with embedding size **d** and head size **d_k**:

Variable	Shape	How it's computed
Input X	(T, d)	Token embeddings, e.g. (6, 768)
W_Q	(d, d_k)	Learned query projection
W_K	(d, d_k)	Learned key projection
W_V	(d, d_k)	Learned value projection
Q = X · W_Q	(T, d_k)	e.g. (6, 64)
K = X · W_K	(T, d_k)	e.g. (6, 64)
Q · K^T	(T, T)	(6, 6) — every token attends to every token
softmax(QK^T/√d_k) · V	(T, d_k)	Weighted values output

GPT-2: **d=768**, **12 heads**, **d_k=64** per head. **12 × 64 = 768** — heads concat back

Multi-Head Attention & Transformer Block

Multi-head: run attention **multiple times in parallel** (e.g., 12 heads in GPT-2).

Each head can learn different relationships:

- Head 1: syntactic (subject-verb)
- Head 2: semantic (noun-pronoun)
- Head 3: positional (nearby words)

One Transformer Block:

Input

- Multi-Head Self-Attention + Residual Connection
- LayerNorm
- Feed-Forward Network (MLP) + Residual Connection
- LayerNorm
- Output

Training — Data, Loss, and Gradient Accumulation

Data: next-token prediction on massive text corpora.

- Input: [The, pizza, came] → Target: [pizza, came, out]
- Loss: **cross-entropy** between predicted and actual next token

Gradient accumulation simulates larger batch sizes on limited GPU memory:

- Instead of 1 batch of 64, do 8 mini-batches of 8
- Accumulate gradients, then update weights once
- Same math, fits in GPU memory

Mixed precision (fp16/bf16): halves memory, doubles speed.

Checkpointing: save model weights periodically to resume if training crashes.

Generation — From Trained Model to Text

After training, generation is **autoregressive** — one token at a time:

```
Prompt:  "The meaning of life is"  
Step 1:  → predict next token → "to"  
Step 2:  "The meaning of life is to" → "find"  
Step 3:  "The meaning of life is to find" → "purpose"  
...
```

Sampling strategies:

- **Greedy:** always pick the highest probability token
- **Temperature:** lower = more deterministic, higher = more creative
- **Top-k:** sample from top k candidates only
- **Top-p (nucleus):** sample from smallest set covering p% probability

nanoGPT implements all of these in ~300 lines of Python.

Source Code Exploration — nanoGPT Locally

`git clone https://github.com/karpathy/nanoGPT` — a whole GPT in ~**600 lines** of Python.

File	What to read	Lines
<code>model.py</code>	<code>CausalSelfAttention</code> , <code>Block</code> , <code>GPT</code> — full architecture	~300
<code>train.py</code>	training loop, gradient accumulation, AdamW, DDP	~350
<code>sample.py</code>	autoregressive generation, temperature, top-k	~90
<code>data/shakespeare_char/prepare.py</code>	char-level tokenizer, train/val split	~40
<code>config/train_shakespeare_char.py</code>	hyperparameters: <code>n_layer</code> , <code>n_head</code> , <code>n_embd</code>	~30

- Read `model.py` **top to bottom** with pen and paper — draw the tensor shapes
- Set a breakpoint in `forward()` — inspect `Q` , `K` , `V` shapes match the slide
- Change `n_layer` `6` → `2` and retrain — watch loss get worse

nanoGPT Deep Dive

The key lesson. Train your own GPT.

- nanoGPT training loop, model architecture
- Weight initialization, data pipeline, sampling

Practice: Train nanoGPT on a small dataset. Modify model size. Train on a Chinese corpus. Change the tokenizer.

GPT Training Engineering

Run real GPT training at scale.

- GPU memory optimization, gradient accumulation
- Mixed precision, checkpointing
- Multi-GPU training, dataset scaling

Practice: Train a 100M parameter model. Resume training. Fine-tune a model.

H200 Trial — DigitalOcean GPU Droplet

Config	H200 ×1	H200 ×8
VRAM	141 GB	1.1 TB
vCPU / RAM	24 / 240 GB	192 / 1920 GB
NVMe Scratch	5 TB	40 TB
Price	\$3.44/hr	\$27.52/hr

- **Linear scaling, no bulk discount** — ×8 is just 8 × ×1
- Rule: **×1** for experiments & fine-tuning, **×8** only for distributed training
- At \$82/day per GPU, every idle hour hurts

Training Setup — FineWeb 60GB

- **Dataset:** FineWeb (HuggingFace) — 60 GB text, ~10B token sample (full = 1.4T)
- **VPS → GPU Droplet transfer:** 5 hours over internet → **5 minutes on private network** (450 MB/s internal)
- **Preprocessing:** 12 parallel CPU workers, ~40M tokens/sec, done in ~20 min
- **Training target:** GPT-2 124M → 1.5B with nanoGPT
- For a 1.5B model to be decent: needs ~100B–1T tokens

Lesson: **never transfer big datasets over public internet** — use the provider's private network.

The 21-Second Iteration — Random Disk Access

Each iteration was **21 seconds** on H200 — way too slow.

- Effective batch: `16 × 1024 × 64 (grad_accum) = ~1M tokens/iter`
- With `gradient_accumulation_steps = 64`, the trainer does **64 random disk reads per iter**
- Data on boot disk, not scratch NVMe → **I/O bottleneck, not compute**
- MFU reported >100% — artifact (nanoGPT's MFU baseline is A100, not H200)

Fix:

- `grad_accum: 64 → 4`, `batch_size: 16 → 256` (same effective batch, **16× fewer disk reads**)
- Move `train.bin` to local NVMe scratch (`/mnt/scratch`)
- Target after fix: **<1 sec/iter, >100k tokens/sec**

Why Runpod > DigitalOcean for Training

- **Per-second billing** — DO bills per hour (pay for idle minutes)
- **30+ GPU types** — RTX 4090, A100, H100, H200, B200 — DO has only a few
- **Community Cloud** pricing much cheaper than Secure Cloud / DO
- **Bring your own Docker** — no fighting with pre-baked images
- **Sub-minute provisioning**, FlashBoot serverless endpoints
- NVMe local scratch by default — fewer I/O surprises

For a solo researcher doing trial runs, Runpod's flexibility + price usually wins.

LLM Agents (Claude Code / OpenClaw)

Build an OpenClaw-style system.

- Tool calling, agent loop
- Planning and execution
- Memory systems, CLI agent design

Practice: Build a coding agent. Build a CLI automation agent. Build a multi-step reasoning agent.

Claude Code Tips — Prompting

- **input + model = output** — the only equation that matters
- **Precise input** — vague prompts produce vague code; be specific about files, signatures, constraints
- **Paste, don't describe** — copy the code snippet, HTML, or error message directly into the prompt
- **One-shot or few-shot** — show an example output when the pattern is non-obvious

Garbage in, garbage out. The model is fixed — you control the input.

Claude Code Tips — Workflow

- **Clone the repo down and ask** — single source of truth beats screenshots and pasted fragments
- **Categorize errors** — fix one big class at a time (all type errors, then all lint, then all tests)
- **Save inputs and outputs** (prompts and responses) — review the good ones later; your input library compounds
- Treat inputs like code: refine, reuse, version them

lzwjava.github.io — My Blog

A Jekyll blog with **~400 original posts**, enhanced with AI-powered tools:

- **LLM Translation** — auto-translated to **9 languages** (EN, ZH, JA, FR, DE, etc.)
- **Google Cloud TTS** — audio versions of posts
- **XeLaTeX** — PDF generation for offline reading
- **EPUB** — ebook export for all post collections
- **GitHub Actions** — automated build, test, translate, deploy
- **MathJax** — renders math in technical posts
- **Night mode**, RSS feed, bilingual content

Plus **8000+ notes**, **323 Python scripts**.

ww — Cross-Platform CLI Toolkit

github.com/lzwjava/ww — A Python CLI for developer productivity with LLM-powered helpers.

Group	Examples
Git	AI commit messages, squash, diff-tree, classify commits
Note	Create notes with git integration, obfuscate data
PDF	Markdown → PDF, batch pipeline, code → PDF
Image	Crop, remove background, compress, screenshots
Search	Multi-engine web search (Bing, DuckDuckGo, Ecosia)
Copilot	OAuth auth, model listing, chat with Copilot API
System	macOS/Linux utils, network scan, process management
Sync	Claude settings, bashrc, ssh configs

iclaw — AI Coding in Restricted Environments

github.com/lzwjava/iclaw — A minimal terminal REPL for AI-assisted coding via GitHub Copilot.

Built for **enterprise-constrained environments** — no browser extensions, no IDE plugins, just a Python CLI.

- **Multi-turn conversations** with Copilot (GPT-5.2 default)
- **Native tool calling** — web search, shell execution, file editing
- **Multiple search providers** — DuckDuckGo, Startpage, Bing, Tavily
- **GitHub OAuth** device flow authentication
- Slash commands: `/model`, `/search`, `/compact`, `/export`, `/copy`

When full-featured tools are blocked, iclaw gets the job done.

Disclaimer: All tests with Copilot were performed using my personal account.

Beyond Engineering

- **Self-taught researcher** — 3 papers on natural vision restoration
- **~400 blog posts** — translated to Chinese with LLMs, ~8000 AI answer notes
- **Life hacker** — hundreds of small innovative practices
- **Traveler** — USA (twice), Hong Kong, Macao, half of China
- English learned from **60+ Filipino teachers** online
- **Crypto & US stock investor** since 2018
- Owns **hundreds of gadgets** — EMF meter, telescope, etc.

Reversing Myopia — 3 Years of Self-Experimentation

- Inspired by **Todd Becker** and **Yin Wang**. 3 papers published. 3 years of data.
- Myopia worsens because we use **full-prescription glasses** for close-up work
- Core principle: **"Just barely clear"** — wear glasses $\sim 1.50\text{D}$ below full prescription
- Left eye: **350** $\rightarrow$ **250** (2022–2024), Right eye improving too

Categorization: separate glasses for **near** (phone, laptop) and **far** (driving) — two glasses for two major scenarios.

Blog post: [Vision Tips](#)

Myopia Experiment — Photos



Let's Connect

Email: lwjava@gmail.com

GitHub: github.com/lwjava

Blog: lwjava.github.io

WeChat: lwjava

Thank You

lwjava@gmail.com · lwjava.github.io

WeChat



Zhiwei Li

Guangzhou, Guangdong



Scan the QR code to add me as a friend.

LinkedIn

linkedin.com/in/lzwjava

or search **Zhiwei Li AI Engineer**